

Questions

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Q1 - 2024 (08 Apr Shift 1)

If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80 (\tan^2 x - \cos x)$ is equal to

- (1) 108
- (2) 109
- (3) 18
- (4) 19

Q2 - 2024 (08 Apr Shift 2)

If the value of $\frac{3 \cos 36^\circ + 5 \sin 18^\circ}{5 \cos 36^\circ - 3 \sin 18^\circ}$ is $\frac{a\sqrt{5}-b}{c}$, where a, b, c are natural numbers and $\gcd(a, c) = 1$, then $a + b + c$ is equal to :

- (1) 40
- (2) 52
- (3) 50
- (4) 54

Q3 - 2024 (09 Apr Shift 1)

Let $|\cos \theta \cos(60 - \theta) \cos(60 + \theta)| \leq \frac{1}{8}$, $\theta \in [0, 2\pi]$. Then, the sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value, is :

- (1) 15π
- (2) 18π
- (3) 6π
- (4) 9π

Questions

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Answer Key

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Solutions

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Q1

$$\sin x = \frac{-3}{5}, \pi < x < \frac{3\pi}{2}$$

$$\tan x = \frac{3}{4} \cos x = -\frac{4}{5}$$

$$80 (\tan^2 x - \cos x)$$

$$= 80 \left(\frac{9}{16} + \frac{4}{5} \right) = 45 + 64 = 109$$

Q2

$$\frac{\frac{3(\sqrt{5}+1)}{4} + 5 \left(\frac{\sqrt{5}-1}{4} \right)}{5 \left(\frac{\sqrt{5}+1}{4} \right) - 3 \left(\frac{\sqrt{5}-1}{4} \right)} = \frac{8\sqrt{5}-2}{2\sqrt{5}+8}$$

$$= \frac{4\sqrt{5}-1}{\sqrt{5}+4} \times \frac{\sqrt{5}-4}{\sqrt{5}-4}$$

$$= \frac{20 - 16\sqrt{5} - \sqrt{5} + 4}{-11}$$

$$= \frac{17\sqrt{5}-24}{-11}$$

$$\Rightarrow a = 17, b = 27, c = 11$$

$$a + b + c = 52$$

Q3

We know that

$$(\cos \theta) (\cos(60^\circ - \theta)) (\cos(60^\circ + \theta)) = \frac{1}{4} \cos 3\theta$$

$$\text{So equation reduces to } \left| \frac{1}{4} \cos 3\theta \right| \leq \frac{1}{8}$$

$$\Rightarrow |\cos 3\theta| \leq \frac{1}{2}$$

$$\Rightarrow -\frac{1}{2} \leq \cos 3\theta \leq \frac{1}{2}$$

$$\Rightarrow \text{maximum value of } \cos 3\theta = \frac{1}{2}, \text{ here}$$

$$\Rightarrow 3\theta = 2n\pi \pm \frac{\pi}{3}$$

$$\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}$$

As $\theta \in [0, 2\pi]$ possible values are

$$\theta = \left\{ \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9} \right\}$$

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Solutions

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Whose sum is

$$\frac{\pi}{9} + \frac{5\pi}{9} + \frac{7\pi}{9} + \frac{11\pi}{9} + \frac{13\pi}{9} + \frac{17\pi}{9} = \frac{54\pi}{9} = 6\pi$$

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